Chapter 96

Deep Image Nodes

This chapter describes Fusion's Deep Image Nodes for compositing objects in 3D space.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Deep Image Compositing Toolset
(Studio Version Only)

Fusion supports a subset of tools to load, utilize, and save deep image data. Deep image data, particularly in formats like OpenEXR, stores multiple samples per pixel to represent data for objects at different depths in the same pixel. The Fusion Deep Image toolset can utilize this valuable information to layer complex 3D renders, particularly where multiple objects overlap, and integrate them with rendered 3D scenes.

The tools can be found in the Deep Image subcategory with a d-prefix.

Additionally:

- Renderer3D nodes can output deep image data.
- Loader and the MediaIn node can import deep image data EXRs.
- Saver can export deep image EXRs.
- All deep compositing tools (outside of Image to Deep and Deep to Image) can automatically convert 2D image inputs to a deep image.

NOTE: Deep compositing requires a linear colorspace. Once merged, users can convert the results into their desired colorspace.

dCrop [dCr]

The dCrop node

dCrop Node Introduction

This tool crops Deep Image samples using specified min and max depth values. It also crops pixels with an input mask.

Inputs

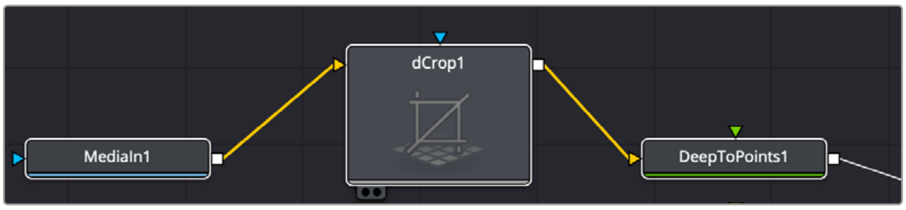
The dCrop tool uses one input.

Input: This orange input is the only required connection. It is connected from a node with deep image data.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools.

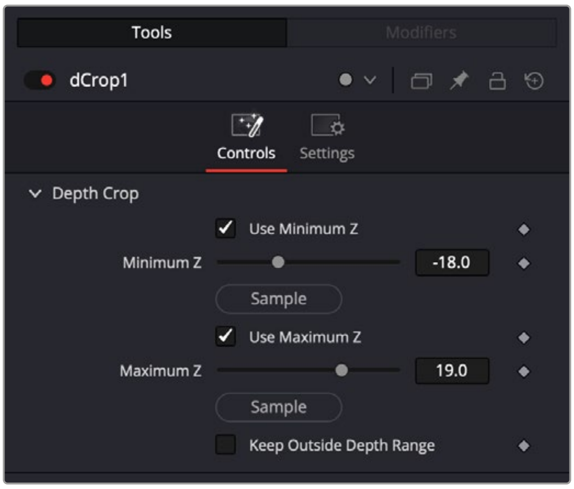
Basic Node Setup

The dCrop node receives the media in containing the deep image data. It then lets you refine the depth boundaries with which to pass to the next tool.



dCrop accepts deep image data

Inspector



dCrop controls

Controls Tab

The Controls tab includes parameters for adjusting the amount of depth information to include or exclude from being passed on the the rest of the node tree.

Depth Crop

Lets you crop the Z parameters of the deep image data.

- **Use Minimum Z:** Check this box to use a Minimum Z depth for the crop.
- **Use Minimum Z:** Adjust the Minimum Z amount using this slider.
- **Sample:** Click and drag from the sample button into the Viewer to choose a value.
- **Use Maximum Z:** Check this box to use a Maximum Z depth for the crop.
- **Keep Outside Depth Range:** Use this control to invert the cropped selection. After setting a minimum and/or maximum value in the sliders above, this control flips the selection so that the cropped region is now visible.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Image nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Deep to Image [DtI]

The Deep to Image node

Deep to Image Node Introduction

This tool converts deep data into a 2D image by flattening its depth samples. This outputs the deep composite ready for traditional 2D compositing.

Inputs

The Deep to Image tool uses one input.

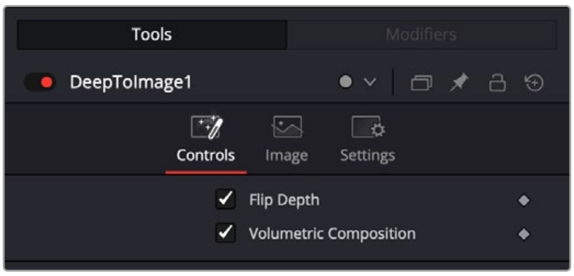
Input: This orange input is the only required connection. It is connected from a node with deep image data.

Basic Node Setup

The Deep to Image node receives the media in containing the deep image data. It then flattens the image for further use in normal 2D compositing.

Deep to Image accepts deep image data, flattens it, then sends the 2D image down the node tree.

Inspector



Deep to Image controls

Controls Tab

The Controls tab includes parameters for adjusting the depth information.

- **Flip Depth:** Inverts the depth information.

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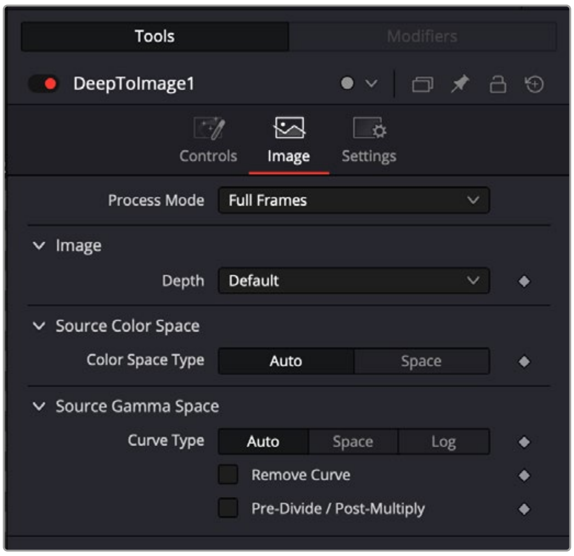
- **Volumetric Composition:** The Volumetric Composition option blends overlapping deep samples as semi-transparent volumes rather than flat surfaces, simulating accurate interaction through elements such as fog or VDBs. This results in a more realistic compositing of semi-transparent elements.

Image Tab

The controls in this tab are used to set the resolution, color depth, and pixel aspect of the rendered image produced by the node.



The Deep to Image Image tab

Process Mode

Use this menu control to select the Fields Processing mode used by Fusion to render changes to the image. The default option is determined by the Has Fields checkbox control in the Frame Format preferences.

Depth

The Depth menu is used to set the pixel color depth of the particles. 32-bit pixels require 4X the memory of 8-bit pixels but have far greater color accuracy. Float pixels allow high dynamic range values outside the normal 0…1 range for representing colors that are brighter than white or darker than black.

Source Color Space

You can use the Source Color Space menu to set the Color Space of the footage to help achieve a linear workflow. Unlike the Gamut tool, this doesn't perform any actual color space conversion, but rather adds the source space data into the metadata, if that metadata doesn't exist. The metadata can then be used downstream by a Gamut tool with the From Image option, or in a Saver, if explicit output spaces are defined there. There are two options to choose from:

- **Auto:** Automatically reads and passes on the metadata that may be in the image.
- **Space:** Displays a Color Space Type menu where you can choose the correct color space of the image.

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Source Gamma Space

Using the Curve type menu, you can set the Gamma Space of the footage and choose to remove it by way of the Remove Curve checkbox when working in a linear workflow. There are three choices in the Curve type menu:

- **Auto:** Automatically reads and passes on the metadata that may be in the image.
- **Space:** Displays a Gamma Space Type menu where you can choose the correct gamma curve of the image.
- **Log:** Brings up the Log/Lin settings, similar to the Cineon tool.

For more information, *see Chapter 99, "Film Nodes,"* in the DaVinci Resolve Reference Manual or Chapter 39 in the Fusion Reference Manual.

Remove Curve

Depending on the selected Gamma Space or on the Gamma Space found in Auto mode, the Gamma Curve is removed from, or a log-lin conversion is performed on, the material, effectively converting it to a linear output space.

Pre-Divide/Post-Multiply

Lets you convert "straight" Alpha channels into pre-multiplied Alpha channels, when necessary.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Image nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Deep to Points [DtP]

The Deep to Points node

Deep to Points Node Introduction

This tool converts Deep images into a 3D point cloud. This tool is useful for visualizing depth samples and offers a way to see elements in Z when incorporating elements.

Inputs

The Deep to Points tool uses one input.

Input: This orange input is the only required connection. It is connected from a node with deep image data.

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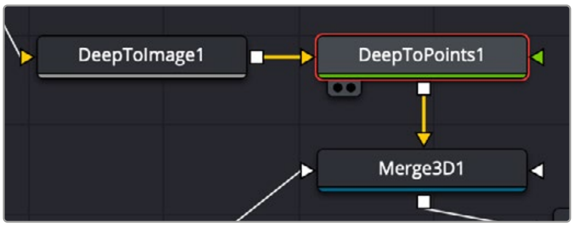
Camera: Connecting a scene's camera to the green camera input precisely positions the point cloud relative to the original 3D scene. The camera's perspective is taken into account so that points are accurately placed within the 3D Viewer.

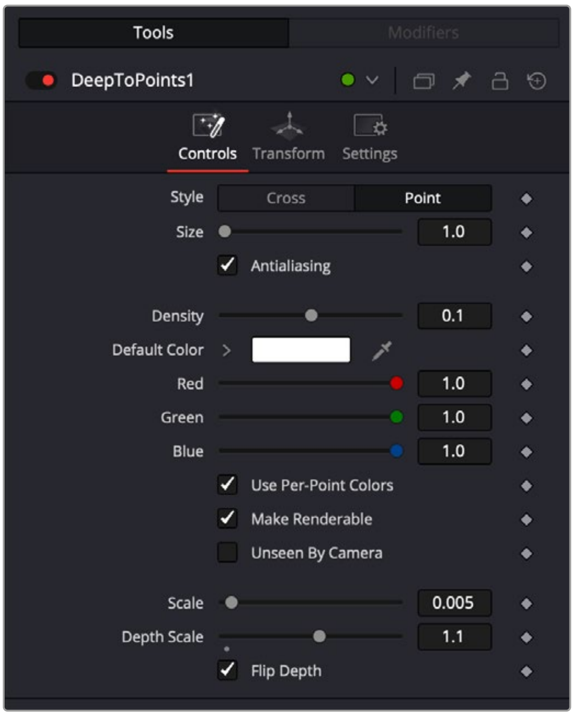
Basic Node Setup

The Deep to Points node receives the media in containing the deep image data. It then creates a 3D point cloud from the image to make it easier to visualize elements in Z space.



Deep to Points accepts deep image data and creates a Point Cloud before passing that visualization off to the rest of the node tree.

Inspector



Deep to Points controls

Controls Tab

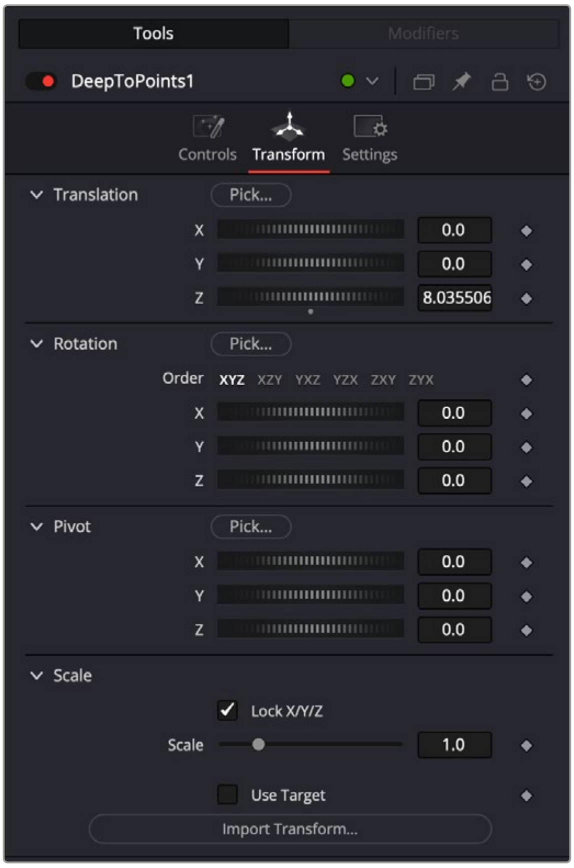
The Controls tab includes parameters for adjusting the point cloud.

- **Style:** Style allows you to display the points as cross hairs or points in the viewer.
- **Size:** This slider can be used to increase the size of the onscreen crosshairs or points
- **Antialiasing:** When enabled, points will appear rounded and smooth, resulting in softer edges.

- **Density:** This slider allows you to control the overall concentration of points in the viewer. Increasing the value will build up the intensity of points whereas decreasing this value will have the opposite effect.
- **Default Color:** When unchecking the Use Per-Point Colors checkbox, each point will inherit a color specified in this control.
- **Use Per-Point Colors:** When enabled, each point will inherit its color from the connected source clip.
- **Make Renderable:** Determines whether the point cloud visible in the 3D viewer will be renderable in a Renderer3D node. Enabling this checkbox will allow the point cloud to be rendered out into a flat 2D image.
- **Unseen by Camera:** This checkbox control appears when the Make Renderable option is selected. If the Unseen by Cameras checkbox is selected, the point cloud is visible in the viewers but not rendered into the output image by the Renderer 3D node.
- **Scale:** This slider allows you to increase or decrease the overall size of the point cloud.
- **Depth Scale:** The slider allows you to expand or contract the overall depth of the point cloud. Increasing the depth scale will amplify the overall distance of depth; spacing out the points giving you a fuller representation of the point cloud.
- **Flip Depth:** Use this checkbox to invert the depth data, flipping the position of the points from negative to positive

Transform Tab

The controls in this tab are used to set the position and scaling parameters of the point cloud.



The Deep to Points Transform tab

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Translation

X, Y, Z Offset

These controls can be used to position the point cloud.

Rotation

Rotation Order

Use these buttons to select which order is used to apply rotation along each axis of the object. For example, XYZ would apply the rotation to the X axis first, followed by the Y axis and then finally the Z axis.

X, Y, Z Rotation

Use these controls to rotate the object around its pivot point. If the Use Target checkbox is selected, then the rotation is relative to the position of the target; otherwise, the global axis is used.

Pivot

X, Y, Z Pivot

A Pivot point is the point around which an object rotates. Normally, an object rotates around its own center, which is considered to be a pivot of 0,0,0. These controls can be used to offset the pivot from the center.

Scale

X, Y, Z Scale

If the Lock X/Y/Z checkbox is checked, a single Scale slider is shown. This adjusts the overall size of the object. If the Lock checkbox is unchecked, individual X, Y, and Z sliders are displayed to allow individual scaling in each dimension. Note: If the Lock checkbox is checked, scaling of individual dimensions is not possible, even when dragging specific axes of the Transformation Widget in scale mode.

Use Target

Selecting the Use Target checkbox enables a set of controls for positioning an XYZ target. When target is enabled, the object always rotates to face the target. The rotation of the object becomes relative to the target.

Import Transform

Opens a file browser where you can select a scene file saved or exported by your 3D application. It supports the following file types:

LightWave Scene	`.lws`
Max Scene	`.ase`
Maya Ascii Scene	`.ma`

The Import Transform button imports only transformation data. For 3D geometry, lights, and cameras, consider using the File > FBX Import option.

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Onscreen Transformation Controls

Most of the controls in the Transform tab are represented in the viewer with onscreen controls for transformation, rotation, and scaling. To change the mode of the onscreen controls, select one of the three buttons in the toolbar in the upper left of the viewer. The modes can also be toggled using the keyboard shortcut Q for translation, W for rotation, and E for scaling. In all three modes, individual axes of the control may be dragged to affect just that axis, or the center of the control may be dragged to affect all three axes.

The scale sliders for most 3D tools default to locked, which causes uniform scaling of all three axes. Unlock the Lock X/Y/Z Scale checkbox to scale an object on a single axis only.

Viewer Transform controls

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Image nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

dHoldout [dHld]

The dHoldout node

dHoldout Node Introduction

This tool uses a foreground Deep Image to occlude background Deep Image samples, either by removing them or computing the value after occlusion

Inputs

The dHoldout tool uses one background input and a foreground input.

Background: This orange input is connected from a node with deep image data that you want to use for the background.

Foreground: The white foreground input is for the second of two deep images which is typically a foreground subject that should be used to occlude the background.

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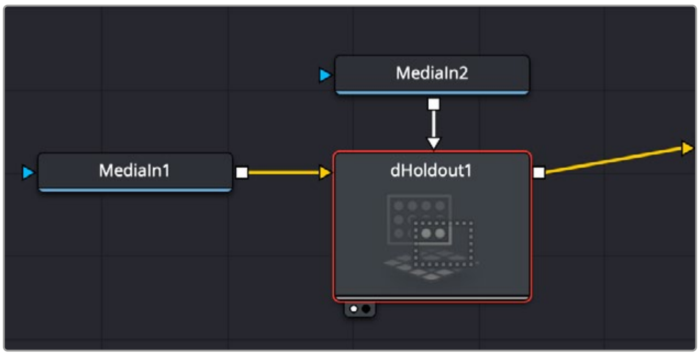
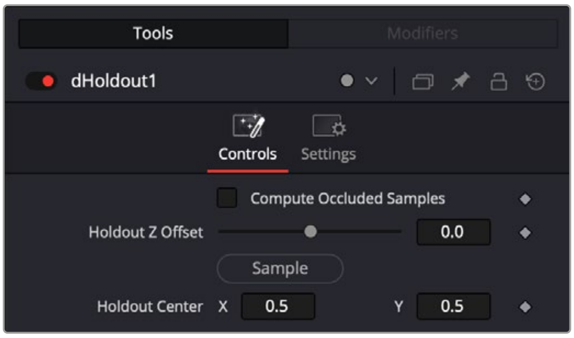
Basic Node Setup

The dHoldout node receives the background media containing the deep image data. It then lets you refine the depth boundaries of the foreground deep image to occlude the background.



dHoldout accepts deep image data from two sources and uses the foreground source to occlude the background.

Inspector



dHolodout controls

Controls Tab

The Controls tab includes parameters for adjusting the amount of depth information to include or exclude from occluding the background deep image.

- **Compute Occluded Samples:** Use this checkbox to remove samples from a semi-transparent foreground input. Visible background pixels will be recolored as if they were seen through a partially transparent foreground.
- **Holdout Z Offset:** This slider allows you to reposition the holdout depth samples forward or backwards. Decreasing this value will push the holdout forward, whereas increasing the value will have the opposite effect.
- **Sample:** Click and drag from the sample button into the viewer to choose a value.
- **Holdout Center:** This allows you to reposition the holdout along the the X and Y axis.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Image nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

dMerge [dMg]

The dMerge node

dMerge Node Introduction

This tool combines multiple data streams, merging their depth samples for seamless compositing. This tool has numerous inputs similar to the merge 3D tool.

Inputs

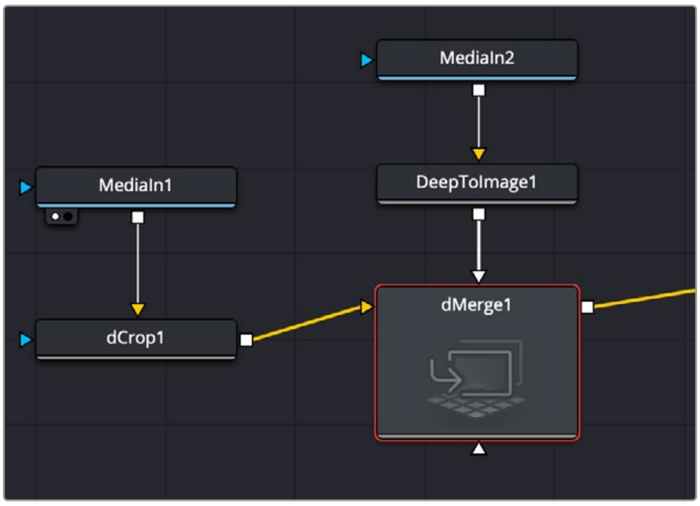
The dMerge tool uses one background input and multiple foreground inputs.

Background: This orange input is connected from a node with deep image data that you want to use for the background.

Foreground: The white foreground input is for connecting other deep image nodes that you want to combine with the background. There can be multiple foreground inputs to this tool just by dragging additional connections between the nodes.

Basic Node Setup

The dMerge node receives the background containing the deep image data. It then lets you merge additional deep image nodes to combine the data.



dMerge accepts deep image data from multiple sources and combines them.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Image nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

dRecolor [dRc]

The dRecolor node

dRecolor Node Introduction

This tool combines Deep Image data with an RGB (multi-layer) image. This node has two inputs, one for Deep/depth and the other for Color. Use this node to recolor Deep Image samples with 2D RGB values

Inputs

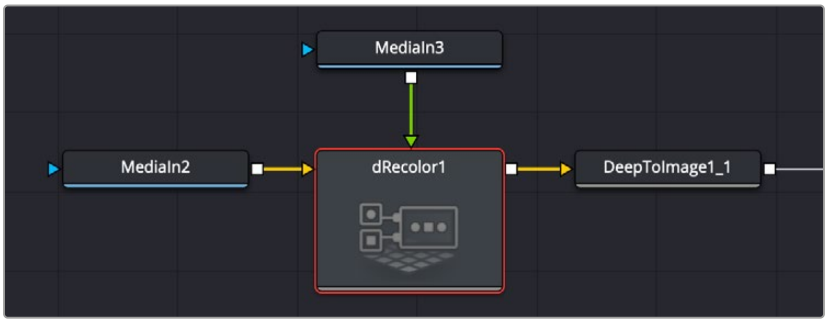
The dRecolor tool uses two inputs, one for deep color images and the other for RGB images.

Depth: This orange input is connected from a node with deep image data that you want to recolor.

Color: The green input is for the RGB image that you want to use to combine the Depth input with.

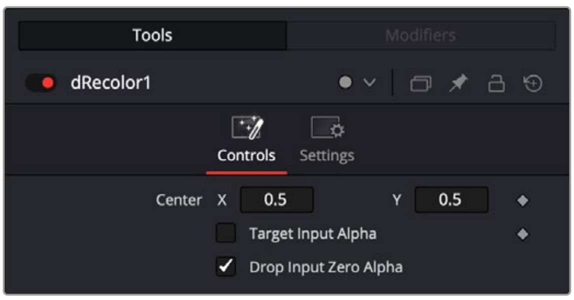
Basic Node Setup

The dRecolor node receives media containing deep image data. It then lets you combine it with a normal 2D image using the RGB input.



dRecolor accepts deep image data and lets you combine it with a 2D RGB image.

Inspector



dRecolor controls

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Controls Tab

The Controls tab includes parameters for blending the 2D RGB image into the background deep image.

- **Center X/Y:** Reposition the 2D RGB image over the deep image.
- **Target Input Alpha:** Select this checkbox to enable the Alpha channel from the Color input, influencing the transparency of deep samples. When this option is turned off, only RGB will be used to recolor the deep image.
- **Drop Input Zero Alpha:** When this is enabled, pixels in the RGBA source with zero alpha are ignored and won't influence the deep image.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Image nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

dResize [dRz]

The dResize node

dResize Node Introduction

This tool resizes a Deep Image's width and height.

Inputs

The dResize tool uses a single input.

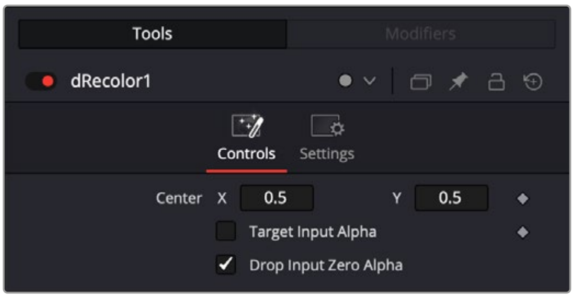
Image: This orange input is connected from a node with deep image data that you want to resize.

Basic Node Setup

The dResize node receives media containing deep image data. It then lets you resize it before passing it onto the rest of the node tree..

dResize accepts deep image data and lets you resize it.

Inspector



dResize controls

Controls Tab

The Controls tab includes parameters for resizing the deep image.

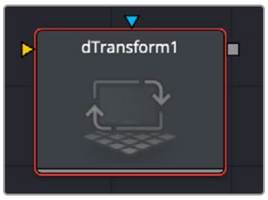
- **Width:** Scales the image's width.
- **Height:** Scales the image's height.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Image nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

dTransform [dXf]



The dTransform node

dTransform Node Introduction

This tool allows you to adjust spatial positioning of deep image data while preserving its depth samples. You can scale and translate Deep Images, including the sample Z values.

Inputs

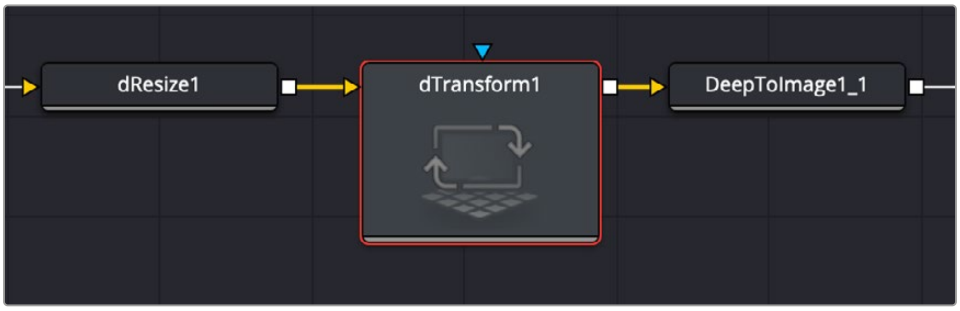
The dTransform tool uses a single input and an effect mask.

Image: This orange input is connected from a node with deep image data that you want to transform.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools.

Basic Node Setup

The dTransform node receives media containing deep image data. It then lets you transform it before passing it onto the rest of the node tree.



dTransform accepts deep image data and lets you transform it.

Inspector

dTransform controls

Controls Tab

The Controls tab includes parameters for transforming the deep image.

- **Scale Z:** The slider allows you to expand or contract the overall depth samples of the source. Increasing the scale will amplify the overall distance of depth. Decreasing will compress the depth range.
- **Translate Z:** Translate Z will allow you to move an object forward or backward along its depth channel. This gives you the control to relocate an object closer or farther away from camera.
- **Sample:** Click and drag from the sample button into the viewer to choose a value.
- **Center X/Y:** Move the center position of the deep image.
- **Pivot X/Y:** Move the pivot point of the deep image.
- **Size X/Y:** Resize the deep image on the selected axis.
- **Use Size and Aspect:** Check this box to activate the parameters below.
- **Size:** Scales the image equally along both the X and Y axes
- **Aspect:** Lets you set the aspect ratio of the image.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Image nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Image to Deep [ItD]

The Image to Deep node

Image to Deep Node Introduction

This tool has an input for a 2D image. When connected, the image is given deep image data and can be composited in a deep environment via the dMerge node.

Inputs

The Image to Deep tool uses one input.

Input: This orange input is the only required connection. It is connected from a node with 2D image data that you want to use with a Deep Image.

Basic Node Setup

The Image to Deep node receives the media in containing the 2D image data. It then adds Deep Image information to it, allowing it to be used in deep image compositing with a dMerge node.

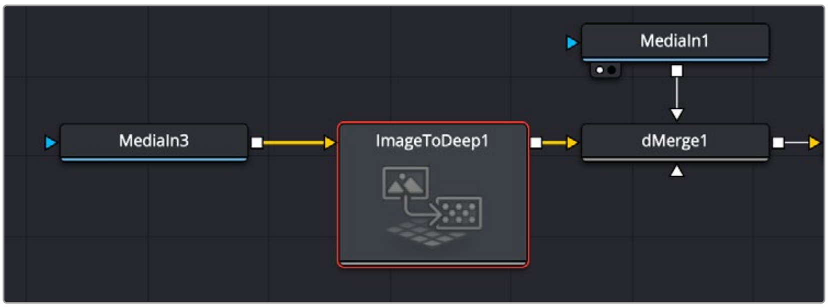


Image to Deep accepts a 2D image (MediaIn3) then adds Deep Image data, allowing it to be composited with Deep Image media (MediaIn1) via a dMerge node.

Inspector

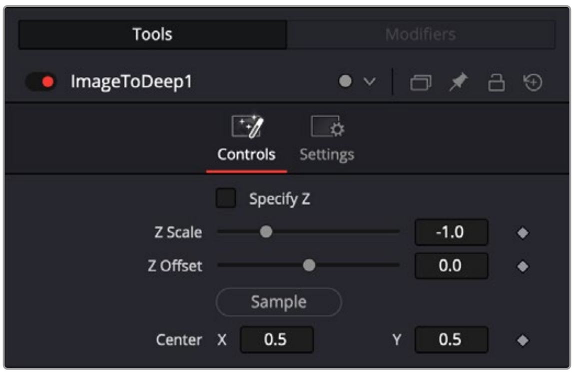


Image to Deep controls

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Controls Tab

The Controls tab includes parameters for adjusting the depth information.

- **Specify Z:** Check this box to specify the current Z depth. If unchecked, you can adjust it using the Z Scale control below.
- **Z Scale:** The slider allows you to expand or contract the overall depth samples of the source. Increasing the scale will amplify the overall distance of depth. Decreasing will compress the depth range.
- **Z Offset:** Will allow you to move an object forward or backward along its depth channel. This gives you the control to relocate an object closer or farther away from camera.
- **Sample:** Click and drag from the sample button into the viewer to choose a value.
- **Center X/Y:** Reposition the 2D image along the X/Y axes.

Common Controls

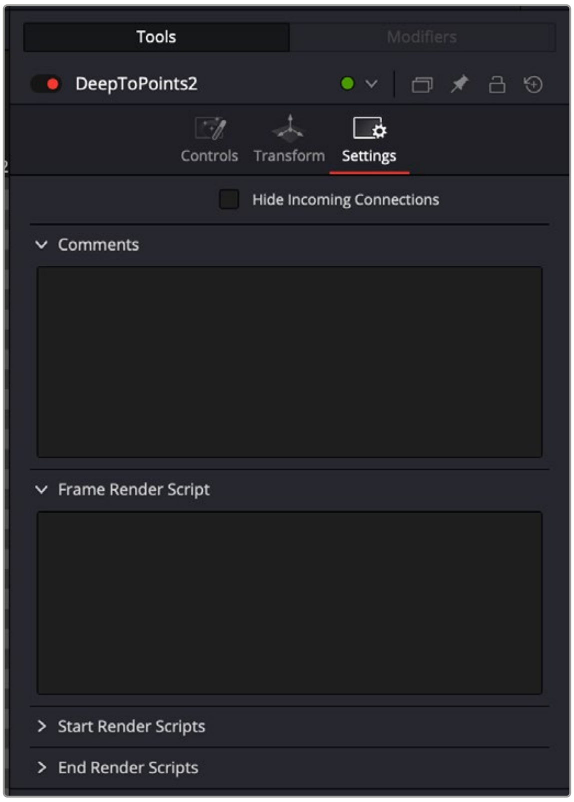
Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Image nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

The Common Controls

Settings Tab

The Common Settings tab can be found on most tools in Fusion. The following controls are specific settings for Deep Image nodes.



The Deep Image Settings tab

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node are displayed in the Inspector. Dragging a connected node from the node tree into the field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

Comments Tab

The Comments tab contains a single text control that is used to add comments and notes to the tool. When a note is added to a tool, a small red dot icon appears next to the setting's tab icon and a text bubble appears on the node. To see the note in the Node Editor, hold the mouse pointer over the node for a moment. The contents of the Comments tab can be animated over time, if required.

Scripting Tabs

The Scripting tab is present on every tool in Fusion. It contains several edit boxes used to add scripts that process when the tool is rendering. For more details on the contents of this tab, please consult the scripting documentation.